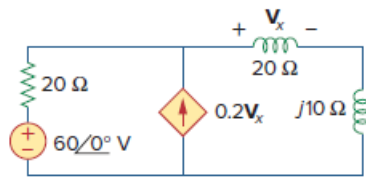


Problem

Determine V_x in Fig. 10.55.

Figure 10.55:



Step-by-step solution

Step 1 of 3

Refer to circuit diagram in Figure 10.55 in the textbook.

Draw the circuit with node notations.

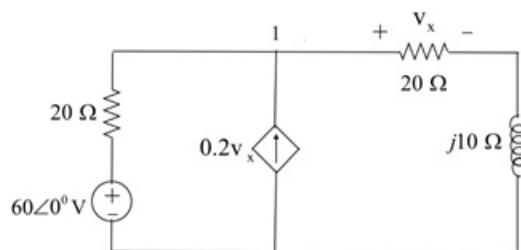


Figure 1

Step 2 of 3

From the circuit, the voltage $\mathbf{V}_x$ is,

$$\mathbf{V}_x = \left(\frac{20}{20 + j10} \right) \mathbf{V}_1 \dots\dots (1)$$

Apply Kirchhoff's current law at node 1.

$$\frac{\mathbf{V}_1 - 60\angle 0^\circ}{20} + \frac{\mathbf{V}_1}{20 + j10} - 0.2\mathbf{V}_x = 0$$

Substitute $\left(\frac{20}{20 + j10} \right) \mathbf{V}_1$ for $\mathbf{V}_x$.

$$\frac{\mathbf{V}_1 - 60\angle 0^\circ}{20} + \frac{\mathbf{V}_1}{20 + j10} - (0.2) \left(\frac{20}{20 + j10} \right) \mathbf{V}_1 = 0$$

$$\left(\frac{1}{20} + \frac{1}{20 + j10} - \frac{4}{20 + j10} \right) \mathbf{V}_1 = \frac{60\angle 0^\circ}{20}$$

$$\left(\frac{1}{20} - \frac{3}{20 + j10} \right) \mathbf{V}_1 = \frac{60\angle 0^\circ}{20}$$

$$\frac{20 + j10 - 60}{(20)(20 + j10)} \mathbf{V}_1 = \frac{60\angle 0^\circ}{20}$$

$$\begin{aligned} \mathbf{V}_1 &= \frac{(60)(20 + j10)}{j10 - 40} \\ &= \frac{120 + j60}{-4 + j} \\ &= -24.7 - j21.176 \text{ V} \end{aligned}$$

Step 3 of 3

Calculate the voltage $\mathbf{V}_x$.

$$\mathbf{V}_x = \left(\frac{20}{20 + j10} \right) \mathbf{V}_1$$

Substitute $-24.7 - j21.176 \text{ V}$ for $\mathbf{V}_1$.

$$\begin{aligned} \mathbf{V}_x &= \left(\frac{20}{20 + j10} \right) (-24.7 - j21.176 \text{ V}) \\ &= (0.8 - j0.4)(-24.7 - j21.176 \text{ V}) \\ &= -28.23 - j7.0608 \text{ V} \\ &= 29.1\angle -166^\circ \text{ V} \end{aligned}$$

Therefore, the value of voltage, $\mathbf{V}_x$ is $\boxed{29.1\angle -166^\circ \text{ V}}$.